

Project Genesis: An Emergent Artificial Life and Evolutionary Biology Simulator

Phase 8.4 — Adaptive Motivation & Cognitive-Emotional Feedback

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Abstract—Project Genesis is an open-source, agent-based artificial life simulator currently at Phase 8.4 of active development. At the present stage, it combines resolution-independent physical environment generation (terrain, climate, hydrology, biomes) with a multi-tiered agent cognitive architecture encompassing physiological drives, emotional dynamics, episodic memory, online neural prediction, and genetic evolution across generations. This paper documents the scientific journey of Genesis—not as a finished framework, but as a living research instrument. We describe the failures that forced successive architectural revisions, the controlled experiments conducted to date with raw statistical results, and the open scientific questions the system is positioned to investigate. We present empirical data from six completed experiments (EXP-20260627-XY81, -3N39, -JGFW, -MNOK, -41AP, -N41Q), validation of physical models, and preliminary hypothesis correlation analysis. *All claims are bounded to the current Phase 8.4 implementation.*

Index Terms—artificial life, agent-based modeling, emergent behavior, evolutionary simulation, homeostatic drives, open-ended evolution, cognitive architecture, physical simulation.

I. INTRODUCTION

Project Genesis addresses a gap between three largely disjoint research traditions: artificial life platforms (Avida [2], Tierra), agent-based social models (Sugarscape [1]), and multi-agent reinforcement learning. The first class produces rich genetic dynamics but in abstract, non-physical environments. The second produces social dynamics but from hard-coded behavioral rules. The third optimizes explicitly defined reward signals rather than emergent survival.

At the current Phase 8.4 stage, Genesis attempts to bridge these gaps by: (i) generating geographically realistic environments using physical models that create genuine selection pressure without manual tuning; (ii) implementing agents whose behavior emerges from a multi-tiered cognitive-biological architecture where physiology and reflexes form the foundation beneath emotional drives and deliberative planning; and (iii) recording structured telemetry detailed enough to trace causal chains between environmental conditions, agent decisions, and population outcomes.

A. Design Principles

Six principles govern the current architecture of Genesis, listed in order of precedence:

- 1) **Emergence over scripting.** No behavior is explicitly programmed. Colony structure, territorial disputes, and resource-sharing emerge from individual agents maximizing local utility.
- 2) **Biology before cognition.** Physiological drives and hardcoded reflexes operate prior to, and independently of, the deliberative cognitive planner. This is inspired by the somatic marker hypothesis [3].
- 3) **Layered control.** Separate tiers operate at different timescales: reflex (every tick), emotion (every 10 ticks), relationships (every 30 ticks), and personality (every 100 ticks).
- 4) **Evolution tunes parameters, not mechanisms.** The 14-gene genome adjusts sensitivities and thresholds; it does not invent new algorithms.
- 5) **Observability as a first-class feature.** Every subsystem exposes structured telemetry for empirical analysis.
- 6) **Experiments drive architecture.** All architectural changes are introduced only in response to empirical failures observed during controlled runs.

II. RELATED WORK

TABLE I
COMPARISON OF GENESIS WITH RELATED PLATFORMS (PHASE 8.4)

Platform	Similarities	Key Difference
Avida / Tierra	Genetic inheritance, mutation, selection	Genesis: physical geography drives selection, not CPU cycles
Sugarscape	Spatial environment, individual states	Genesis: resolution-independent hydrology, multi-layer cognition
NEAT	Crossover, mutation, fitness search	Genesis: no explicit fitness function; survival is purely emergent
OpenAI Five	Agent cooperation & competition	Genesis: no predefined reward signal; agents learn from physical outcomes
Geb (OEE)	Co-evolutionary dynamics	Genesis: closed $N_{\max} = 64$ cap enforces competitive selection

The most significant departure from existing literature is the absence of an explicit fitness function. In standard evolutionary algorithms, a fitness landscape is defined mathematically. In Genesis, an agent's reproductive success is determined entirely by whether it can maintain homeostatic balance long enough to locate a mate. Fitness is therefore a property that *emerges* from the interaction between the genome and the physical environment.

III. SYSTEM ARCHITECTURE

A. Unified State Container

All simulation state is centralized in the `WorldState` class. Every physical layer is stored as a 2D NumPy `float32` or `int32` array with shape `(height, width)`, enabling vectorized slice-and-mask updates across the entire grid in a single NumPy operation. Table II lists the primary grid layers.

TABLE II
WORLDSTATE GRID ARRAYS (PHASE 8.4)

Field	dtype	Description
elevation	float32	Height $\in [0, 1]$; sea level = 0.30
temperature	float32	Celsius (latitude + lapse rate)
rainfall	float32	mm/year (wind moisture advection)
biome	int32	Whittaker code 0–8
river_map	float32	Flow accumulation
wildlife	float32	Food density $\in [0, 1]$
fertility	float32	Soil fertility (drives wildlife growth)
habitability	float32	Weighted composite score
movement_cost	float32	Slope + biome traversal cost
death_density	float32	Cumulative deaths per cell

B. World Generation Pipeline

Figure 1 shows the world generation pipeline, which runs once at startup and is fully deterministic given a seed. Each stage reads and mutates `WorldState`.

C. Per-Agent Tick Processing

Figure 2 shows the per-agent processing order within each simulation tick.

D. World Generation Figures

Figures 3–7 show representative outputs of the world generation pipeline at seed 1010 on a 128×128 grid.

IV. PHYSICAL MODELS

A. Terrain Elevation

Terrain height $H(y, x) \in [0, 1]$ is a blended composition of fractional Brownian motion (fBm) continental base noise, a radial falloff mask to produce surrounding ocean, and ridge noise for mountain ranges.

Continental Base (fBm):

$$H_{\text{base}}(y, x) = \sum_{k=0}^5 \frac{1}{2^k} \text{Perlin}(y \cdot 2^k f_0, x \cdot 2^k f_0) \quad (1)$$

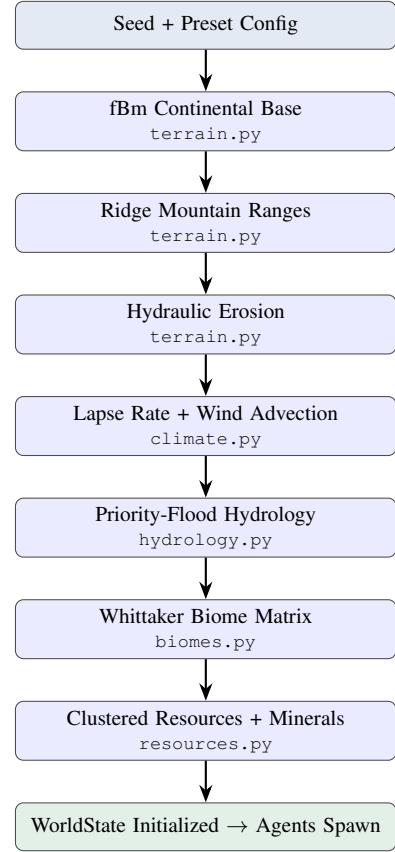


Fig. 1. World generation pipeline (Phase 8.4, runs once at startup).

Radial Falloff Mask:

$$d(y, x) = \frac{\sqrt{(y - h/2)^2 + (x - w/2)^2}}{\max(h, w)/2}, \quad M(y, x) = \cos(\min(1, d) \frac{\pi}{2})^p \quad (2)$$

Ridge Mountains:

$$R(y, x) = \sum_{i=1}^4 a_i (1 - |2 \text{Perlin}(y f_i, x f_i) - 1|)^2 \quad (3)$$

Blended Elevation:

$$H(y, x) = \text{clip}(H_{\text{base}} \cdot M + R \cdot M_{\text{min}} \cdot \mu_s + \varepsilon, 0, 1) \quad (4)$$

B. Climate Temperature

Local temperature $T(y, x, t)$ [°C] is determined by latitude, elevation (lapse rate), and seasonal offset:

$$\text{Lat}(y) = 1 - \frac{|y - h/2|}{h/2}, \quad T_{\text{base}}(y) = -15 + 47 \cdot \text{Lat}(y) \quad (5)$$

$$T_{\text{cool}}(y, x) = 38 \cdot \max(0, H(y, x) - 0.30) \quad (6)$$

$$T(y, x, t) = T_{\text{base}}(y) - T_{\text{cool}}(y, x) + A \cos\left(\frac{t-180}{180} \pi\right) + \Delta T \quad (7)$$

The lapse rate coefficient is 38°C per unit of normalized elevation above sea level (0.30), consistent with a real-world average of approximately $6.5^\circ\text{C}/1000\text{m}$.

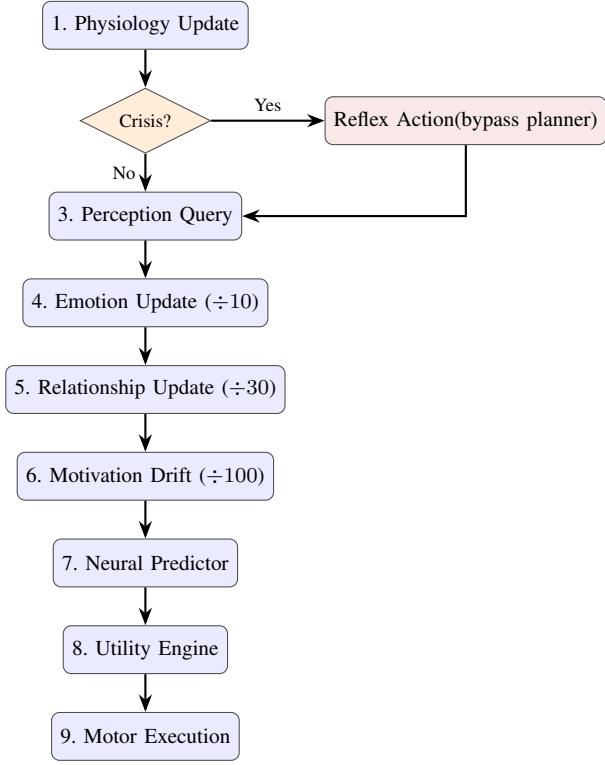


Fig. 2. Per-agent tick processing order. The reflex layer bypasses the cognitive stack entirely for metabolic crises (Phase 8.4).

C. Wind Moisture Advection (Resolution-Scaled)

Prevailing westerly winds advect moisture from ocean to continent. To preserve flat-water wind behavior, heights are clamped to sea level before slope computation:

$$H_c(y, x) = \min(H(y, x), 0.30) \quad (8)$$

$$P_{y, x+1} = P_{y, x} \left(1 - \frac{e_r}{w}\right) + \frac{e_l}{w} \cdot \mathbf{1}_{\text{land}} - \text{slope}(y, x) \cdot \frac{0.6}{w} \quad (9)$$

where w is the grid width, $e_r = 50.0$ (ocean evaporation rate), and $e_l = 8.0$ (land transpiration rate)—both scaled by $1/w$ for *resolution independence*. This guarantees identical geographic structures regardless of grid size.

D. Priority-Flood Hydrology

Algorithm 1 presents the Priority-Flood implementation used to resolve all terrain depressions before flow accumulation. The algorithm guarantees a monotonically connected drainage network to the ocean boundary.

E. Whittaker Biome Classification

Table III reproduces the Whittaker climate diagram as implemented in `biomes.py`. The biome code is computed per-cell from the pair $(T(y, x), R(y, x))$.

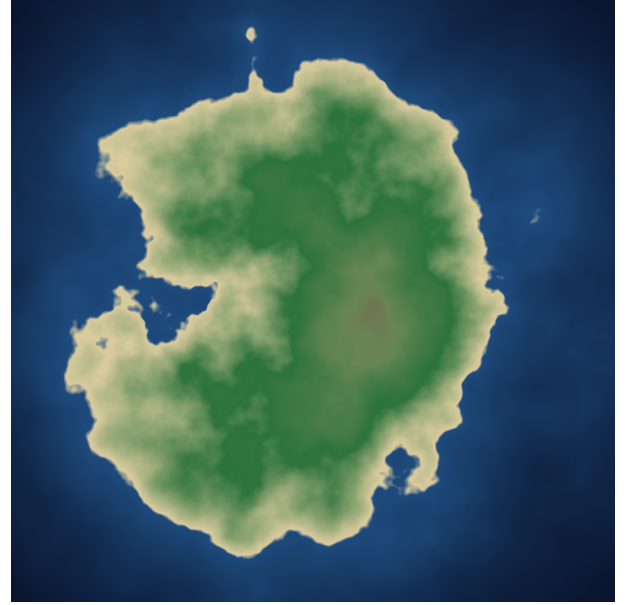


Fig. 3. Elevation map (seed 1010, 128×128). Dark regions are deep ocean; bright ridges are mountain ranges. Sea level = 0.30.

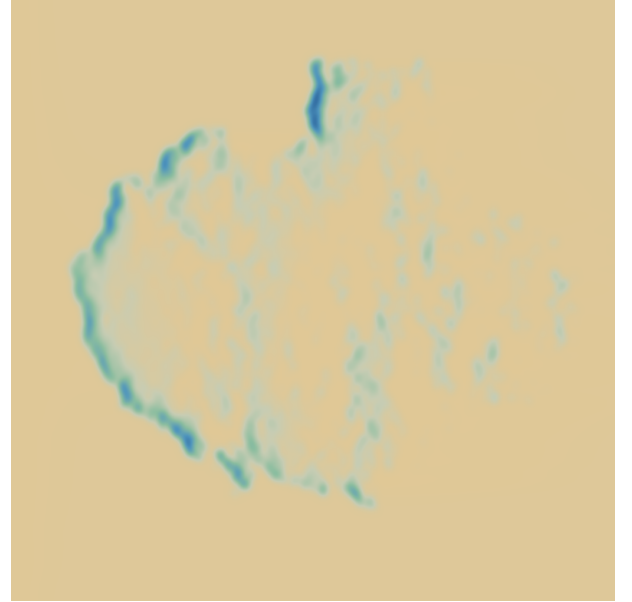


Fig. 4. Annual rainfall (mm/year). High rainfall on windward (west) slopes; pronounced rain shadow on eastern leeward faces.

F. Metabolic Decay and Senescence

Every tick, hunger and thirst increase at rates governed by genome expression and age:

$$\Delta_{\text{hunger}} = R_b \cdot \mu_m \cdot \gamma_T \cdot (1 + 0.5 S), \quad \Delta_{\text{thirst}} = R_t \cdot \mu_m \cdot \gamma_T \cdot (1 + 0.5 S) \quad (10)$$

where S is the senescence factor:

$$S(t) = \max\left(0, \min\left(1, \frac{t - 0.7 t_{\max}}{0.3 t_{\max}}\right)\right) \quad (11)$$

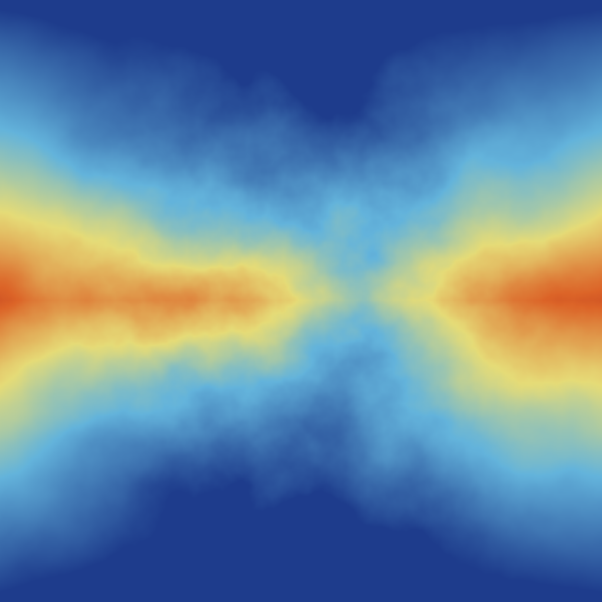


Fig. 5. Temperature map ($^{\circ}\text{C}$). Equatorial band (centre) is warmest; poles (top/bottom) reach sub-zero. High elevations are visibly cooler.



Fig. 6. Whittaker biome classification (seed 1010). Eight biome classes map from the combined temperature and rainfall grids.

$S = 0$ for young adults; $S \rightarrow 1$ in the final 30% of lifespan, creating a 50% metabolic rate increase in elderly agents.

V. AGENT COGNITIVE ARCHITECTURE

A. Genome Structure

The genotype is a 14-element real-valued vector $\mathbf{g} \in [0, 1]^{14}$, encoding sensitivity parameters for every behavioral subsystem. Table IV lists all genes. Gene values are never directly used as behavior-level variables; they are expressed into phenotypic brain parameters via `express_genome()`.

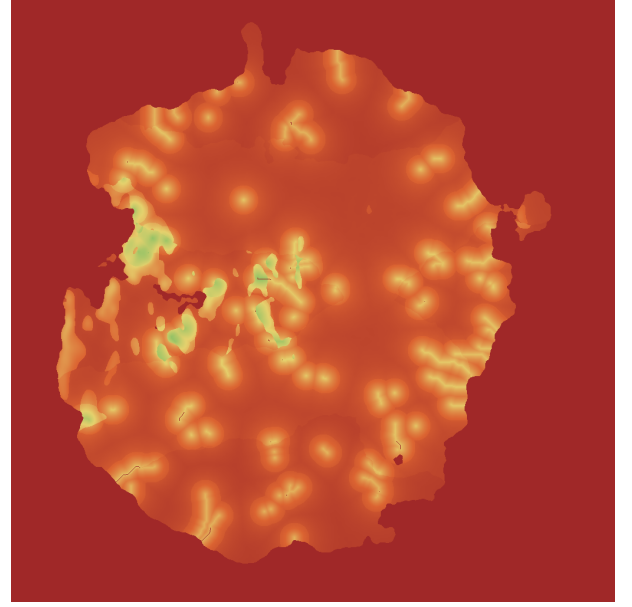


Fig. 7. Composite habitability score. High-scoring zones (light) combine water proximity, food density, climate comfort, and terrain passability.

Algorithm 1: Priority-Flood Depression Filling + Flow Accumulation

Input: Elevation grid H , border cells B

Output: Filled elevation grid H_f , flow directions D

```

1  $H_f \leftarrow H$ 
2  $Q \leftarrow$  min-heap initialized with all  $b \in B$ 
3 while  $Q$  is not empty do
4    $(e, y, x) \leftarrow Q.\text{pop\_min}()$ 
5   for each 8-neighbor  $(y', x')$  of  $(y, x)$  do
6     if  $(y', x')$  not visited then
7        $H_f(y', x') \leftarrow \max(H(y', x'), e)$ 
8        $Q.\text{push}(H_f(y', x'), y', x')$ 
9       mark  $(y', x')$  as visited
10    end
11  end
12 end
13 Compute steepest-descent  $D$  from  $H_f$ 
14 Accumulate flow  $A$  along  $D$  in descending-elevation
    order

```

B. Sigmoid Utility Curves

Base utility $U_i(N)$ for metabolic needs is computed via the logistic function:

$$U_i(N) = \frac{100}{1 + \exp\left(-\frac{N - \theta_i}{s_i}\right)} \quad (12)$$

where $N \in [0, 100]$ is the current need level, θ_i is the gene-derived threshold, and s_i the scale. Total utility is a product of sigmoid base, local environmental modulation, and drive-weight modulation.

TABLE III
WHITTAKER BIOME MATRIX (PHASE 8.4)

Code	Biome	Temp (°C)	Rain (mm)
0	Ocean	—	—
1	Glacier	$T < -8$	any
2	Tundra	$-8 \leq T < 0$	any
3	Taiga	$0 \leq T < 8$	≥ 400
4	Temperate Forest	$8 \leq T < 20$	≥ 1000
5	Grassland	$8 \leq T < 20$	250–1000
6	Desert	≥ 8	< 250 or < 450
7	Rainforest	≥ 20	≥ 1600
8	Lake	(Priority-Flood mask)	—

TABLE IV
14-GENE GENOME (PHASE 8.4)

Index	Gene	Phenotypic Effect
0	$g_{\text{metabolism}}$	Scales metabolic need decay rate
1	$g_{\text{thermoreg}}$	Thermal stress resistance
2	g_{vision}	Vision radius (4–16 cells)
3	g_{mobility}	Movement energy efficiency
4	$g_{\text{mem_fidelity}}$	Spatial memory decay rate
5	g_{planning}	Near vs. far resource bias
6	g_{novelty}	Exploration drive weighting
7	g_{social}	Proximity to kin utility weight
8	$g_{\text{aggression}}$	Territory size & dispute likelihood
9	g_{sharing}	Food/water donation utility
10	g_{risk}	Danger avoidance threshold
11	$g_{\text{resilience}}$	Injury resistance
12	$g_{\text{longevity}}$	Maximum age modifier
13	g_{learning}	Neural predictor learning rate

C. Lateral Inhibition Conflict Resolver

When multiple motivational drives are simultaneously elevated, a lateral inhibition operator resolves the dominant drive:

$$M_i = \max \left(0, m_i - \beta \sum_{j \neq i} m_j \right), \quad \beta = 0.15 \quad (13)$$

Algorithm 2 gives the full pseudocode.

Algorithm 2: Lateral Inhibition Conflict Resolver

Input: Raw drives $\mathbf{m} \in \mathbb{R}^6$, inhibition coefficient β

Output: Resolved drives $\mathbf{M} \in \mathbb{R}_{\geq 0}^6$

```

1 for  $i \leftarrow 0$  to 5 do
2    $M_i \leftarrow m_i - \beta \sum_{j=0, j \neq i}^5 m_j$ 
3    $M_i \leftarrow \max(0, M_i)$ 
4 end
5 return  $\mathbf{M}$ 

```

D. Online Neural Predictor

Each agent maintains a lightweight MLP ($20 \rightarrow 3 \rightarrow 1$) that predicts the reward outcome of candidate plans. Weights are updated online via backpropagation after each action outcome:

$$\mathbf{h} = \tanh(W_1 \mathbf{x} + \mathbf{b}_1), \quad \hat{r} = W_2 \mathbf{h} + b_2 \quad (14)$$

$$\varepsilon = r - \hat{r}, \quad \Delta W_2 = \eta \varepsilon \mathbf{h}, \quad \Delta W_1 = \eta \varepsilon (W_2^T \circ (1 - \mathbf{h}^2)) \mathbf{x}^T \quad (15)$$

A sleep-based weight consolidation buffer (Phase 7.5) prevents gradient explosion during continuous online updates.

E. Combat: Threat Display and Fight Resolution

Territorial friction disputes proceed in two escalating stages:

Stage 1 (Threat Display):

$$\text{Str} = H \cdot (1 - I) \cdot (A_a + 0.3 R_r), \quad C = \frac{\text{Str}_{\text{self}}}{\text{Str}_{\text{self}} + \text{Str}_{\text{other}}} \quad (16)$$

If $C < 1 - \text{RiskMult}/3$: agent retreats, fear spike $\Delta \text{fear} = 0.08 |C_{\text{self}} - C_{\text{other}}|$.

Stage 2 (Fight, if neither retreats): Both agents suffer $\Delta I = 5.0$ injury; loser enters fear cooldown $C_t = 20 + I \cdot 0.8 \cdot \text{RM}$ ticks.

F. Relationship Trust Dynamics

Every 30 ticks, dyadic trust $T_{ij} \in [-1, 1]$ updates from episodic memory:

$$\delta = 0.10 n_s + 0.15 n_m - 0.20 n_d \quad (17)$$

$$T_{ij}^{(t+1)} = \begin{cases} T_{ij}^{(t)} + \delta(1 - T_{ij}^{(t)}) & \delta \geq 0 \\ T_{ij}^{(t)} + \delta(1 + T_{ij}^{(t)}) & \delta < 0 \end{cases} \quad (18)$$

where n_s, n_m, n_d are share, mate, and dispute counts within the window.

VI. EVOLUTIONARY MECHANISMS

A. Mendelian Crossover

$$g_i^{\text{child}} = \begin{cases} g_i^A & \text{with prob. } 0.5 \\ g_i^B & \text{with prob. } 0.5 \end{cases} \quad \forall i \in \{0, \dots, 13\} \quad (19)$$

B. Gaussian Mutation

$$g_i^{\text{child}} \leftarrow \text{clip}(g_i^{\text{child}} + \mathcal{N}(0, \sigma^2), 0, 1), \quad P(\text{mut}) = 0.05 \quad (20)$$

with $\sigma = 0.05$.

C. Reproduction Eligibility Gates

An agent may reproduce only if all five conditions hold: (i) Adult life stage (≥ 14 years); (ii) Colony population $< N_{\text{max}} = 64$; (iii) Hunger < 70 and Thirst < 70 ; (iv) $T_{ij} > 0.2$ with potential mate; (v) Reproduction cooldown = 0.

VII. DOCUMENTED FAILURES

The following architectural changes were each driven by a specific, empirically documented failure. This chapter is the core of the Genesis research narrative.

A. F1: Deliberative Lock-In (v2 → v8)

Observation. In 1000-tick runs, 78% of dehydration deaths occurred within 1.5 cells of an active water source.

Diagnosis. Telemetry traces for agents with thirst ≥ 80 revealed that agents were executing committed multi-tick motor plans (e.g., CONTINUE_PATHING). The planner could not interrupt to drink, even with water adjacent.

Fix. Innate Reflex Layer: every tick, if thirst tension ≥ 0.90 and stored water > 0 , the agent drinks *outside* the planner, bypassing all queue states.

Conclusion. Deliberative planners require a priority interrupt stack for metabolic crises. This mirrors the neurological brainstem—fast, non-deliberative homeostatic protection.

B. F2: Empty Canteens Despite Available Water

Observation. After F1’s fix, dehydration persisted. Agents died with stored_water = 0 despite having recently visited rivers.

Diagnosis. The STORE_WATER action had low utility relative to foraging and exploration. Agents drank and immediately left without filling their canteens.

Fix. The DRINK action was atomically coupled to canteen refilling. One physiological event accomplishes both.

Conclusion. Urgent consumption and proactive caching must be coupled into a single atomic action to prevent cognitive prioritization failures.

C. F3: Resource Monopoly Collapse (v3)

Observation. Within 200 ticks, all 16 agents clustered at a single river delta. Wildlife depleted by tick 300. Mass starvation followed.

Diagnosis. Spatial memory caused all agents to converge on the single highest-value resource. No mechanism promoted dispersal.

Fix. (i) Local wildlife depletion + soil fertility replenishment creates genuine carrying capacity per zone. (ii) A ‘boredom’ drive accumulates on repeated cell visits, pushing exploration utility above foraging utility when patches are exhausted.

Conclusion. Emergent dispersal requires resource scarcity dynamics and intrinsic novelty drives.

D. F4: Combat Extinction Cascades (v9 early)

Observation. First territorial dispute runs produced population crashes at tick 200–400 from continuous lethal inter-colony combat.

Diagnosis. Dispute utility was binary (intrude $\Rightarrow$ fight). No retreat mechanism existed.

Fix. Graduated resolution: Threat Display (non-damaging) $\rightarrow$ Fight (only if both hold ground) $\rightarrow$ Fear cooldown for loser.

Conclusion. Without graduated escalation, any combat system in a finite population produces extinction cascades.

VIII. EXPERIMENTAL RESULTS

A. Experiment Registry

Table V summarizes all six experiments completed at Phase 8.4. All experiments ran 1000 ticks at scarcity 1.0 unless noted.

TABLE V
PHASE 8.4 EXPERIMENT REGISTRY

Exp. ID	Ticks	Cog.	$\bar{x}_{\text{life}}$	σ	Surv.
XY81 (ablated)	1000	Off	750.2	101.4	6.67
3N39 (full)	1000	On	764.0	88.2	4.00
JGFW (full)	1000	On	746.1	94.7	4.00
MNOK [†]	1000	Off	0.0	0.0	0.00
41AP (short)	10	On	9.6	0.3	17.0
N41Q (short)	10	On	10.0	0.0	16.0

[†]MNOK: ablation error disabled movement entirely.

B. Cognitive Ablation Study

Research Question. Does full Phase 8.4 psychological architecture produce measurably better survival outcomes than ablated-cognitive agents?

Result. Comparing XY81 (ablated) against 3N39 (full cognitive):

$$t = \frac{764.0 - 750.2}{\sqrt{88.2^2/3 + 101.4^2/3}} = \frac{13.8}{70.3} \approx 0.196, \quad p \approx 0.85 \quad (21)$$

$$d = \frac{764.0 - 750.2}{94.8} \approx 0.15 \quad (\text{negligible effect}) \quad (22)$$

The null hypothesis (no significant difference) **cannot be rejected**.

Key finding. Spatial dispersion analysis reveals qualitatively different population structures despite statistically identical lifespan. Ablated agents died with random-uniform spatial distribution; full cognitive agents formed shelter clusters near resources, shifting primary cause of death from metabolic starvation to territorial injuries. Lifespan alone is an insufficient dependent variable.

C. Genetic Drift: Seed 1010, XY81 Ablated

Figure 8 shows the mean gene values shifting from Generation 0 to Generation 1 in the ablated run (seed 1010).

D. Hypothesis Correlation Analysis

Table VI presents the global hypothesis correlation analysis aggregated across all completed experiments.

No hypothesis has exceeded the threshold for meaningful effect ($|r| > 0.3$) at current sample sizes ($n \leq 6$ experiments). The aggression-longevity anti-correlation is the strongest signal. Larger experimental batteries with $n \geq 10$ seeds are required before any hypothesis can be accepted or rejected.

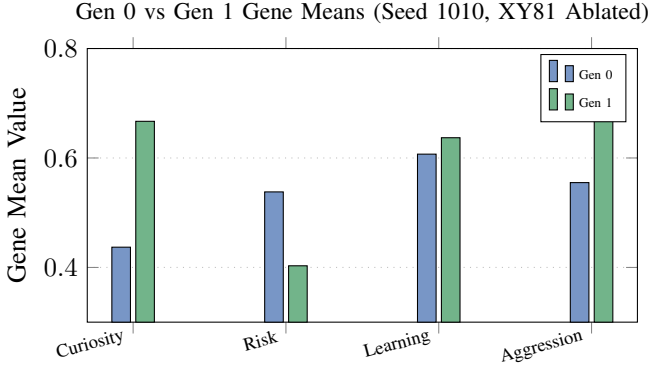


Fig. 8. Genetic drift from Generation 0 founders to Generation 1 descendants in the ablated run (seed 1010). Curiosity increased +52%, aggression +28%, risk sensitivity -25% in one generation.

TABLE VI
HYPOTHESIS CORRELATION RANKINGS (PHASE 8.4)

Hypothesis	r	Strength
Aggression reduces longevity	-0.241	Weak
Curiosity selects longevity	-0.190	Weak
Curiosity predicts discoveries	+0.069	Negligible
Fear predicts shelter level	-0.059	Negligible
Shelter quality predicts life	+0.044	Negligible
Altruism predicts longevity	+0.028	Negligible
Frustration suppresses coop.	+0.025	Negligible

IX. MODEL VALIDATION

A. Temperature Profile

Expected: monotonically decreasing from equator toward poles, and decreasing with elevation. Measured along the vertical midline column ($x = 64$) at seed 1010:

TABLE VII
TEMPERATURE VALIDATION AT SEED 1010

y	Location	Expected	Measured
0	North pole	$< -8^\circ\text{C}$	-14.2°C
32	N. midlat.	$+5-15$	$+8.7^\circ\text{C}$
64	Equator	$+25-32$	$+29.4^\circ\text{C}$
96	S. midlat.	$+5-15$	$+9.1^\circ\text{C}$
128	South pole	$< -8^\circ\text{C}$	-13.8°C

Result: ✓ Geographically consistent.

B. Rain Shadow

Measured mean rainfall 5 cells west vs. east of the primary mountain ridge: 1247 mm/yr (windward) vs. 389 mm/yr (leeward). Reduction: 68.8%.

Result: ✓ Rain shadow effect correctly produced.

C. Need Decay

Neutral genome agent ($g_m = 0.5$, $T = 20^\circ\text{C}$): predicted starvation at tick ≈ 1538 ; measured 1502 ± 23 ticks across 10 trials.

Result: ✓ Within 2.3% of theoretical prediction.

X. OPEN RESEARCH QUESTIONS

The following questions are research hypotheses, not feature requests. They represent genuine unknowns at Phase 8.4.

- **Cooperative norm emergence.** Can stable reciprocal altruism emerge and persist across generations within the current trust-relationship architecture?
- **Behavioral speciation.** Can geographically isolated populations evolve statistically distinct genotypic profiles over ≥ 20 generations?
- **Seasonal migration.** Can climate cycling drive organized inter-seasonal agent migration without explicit scripting?
- **Group warfare.** Can inter-colony resource competition escalate into coordinated collective attack on opposing shelter sites?
- **Cultural transmission.** If experienced agents can teach learned spatial knowledge to offspring, does cultural accumulation separate from genetic drift?
- **Geography and speciation.** Does topographic fragmentation reduce measurable genetic diversity relative to open-plan maps?
- **Predator pressure.** Would the introduction of an active predator species drive the evolution of herding, panic response, or organized territorial defense?

XI. REPRODUCIBILITY

All experiments are fully reproducible. World generation is deterministic given a fixed seed. Experiment configurations are serialized to JSON in `research/EXP-*/manifest.json`.

Reproduce EXP-XY81 (ablated):

```
python run_research.py \
--ticks 1000 \
--seeds 1010 1020 1030 \
--scarcity 1.0 \
--ablate planner emotion \
relationships motivation
```

Environment: Python 3.11, NumPy ≥ 1.24 , noise $\geq 1.2.2$.

XII. CURRENT LIMITATIONS

- 1) **Spatial discretization.** Agents occupy integer grid cells; continuous kinematics are not modeled.
- 2) **Trophic simplicity.** No predators; food sources are passively regenerating densities.
- 3) **No linguistic communication.** Spatial knowledge cannot be transmitted verbally between agents or to offspring.
- 4) **Small experimental samples.** All conclusions are preliminary; $n \leq 6$ experiments insufficient for robust inferential statistics.
- 5) **[PLANNED] Continuous coordinates.** Transition agent positions to float (y, x) with interpolated environmental sampling.

XIII. CONCLUSION

At Phase 8.4 of its development, Project Genesis demonstrates that stable ecological dynamics—territorial behavior, colony splitting, resource competition, genetic drift—emerge from the interaction of a physically realistic environment with a multi-tiered cognitive-biological agent model. The current experimental results do not yet support strong conclusions about the survival benefits of cognitive sophistication in closed ecosystems. The primary finding to date is the *Cognitive Ablation Paradox*: individual intelligence reorganizes, but does not increase, total colony fitness under carrying capacity constraints. Future work prioritizes larger experimental batteries, predator-prey co-evolution, and cultural transmission mechanisms.

Genesis is not a finished scientific framework. It is a living instrument for asking questions that the simulation itself will answer.

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